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JURASSIC PALEOMAGNETIC CONSTRAINTS ON THE COLLISION OF THE NORTH AND SOUTH CHINA BLOCKS

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Abstract. This paper reports a paleomagnetic study of Jurassic rocks from the Ordos Basin in the North China Block (NCB). A recent remagnetization and a high-temperature and/or high-coercivity component with dual polarities have been isolated. The Middle Jurassic pole (74°N, 233°E, A95=5°) is roughly consistent with previous results, and the Lower Jurassic pole (82°N, 286°E, A95=7°) is located between the late Triassic and Middle Jurassic poles. These results, combined with paleomagnetic results from adjacent blocks, suggest that 1) the pole for the NCB suffered an angular displacement of 47°+/- 8° between the late Triassic and Middle Jurassic, 2) the SCB underwent final accretion to the NCB in the Middle Jurassic, and 3) accretion of the NCB to Siberia was not complete until the late Jurassic and possibly even until the early Cretaceous.

Introduction

Paleomagnetism has proved to be a useful tool to unravel the complex sequence of relative motions and collisions of the many blocks that together form the Asian mosaic (e.g., North China Block (NCB), South China Block (SCB), Tarim and the numerous accreted terranes of the Tibetan region)(Figure 1). Late Paleozoic and early Mesozoic formations have been intensively studied by modern paleomagnetic methods since 1981 (e.g. McElhinny et al., 1981). The collision of the NCB and SCB was suggested on that basis to have taken place between the Middle Triassic and the Middle Jurassic (Lin, 1984; Zhao and Coe, 1987). The precise timing of complete suturing was not well constrained due to the scarcity of reliable results for the middle to late Mesozoic and to the major problem of widespread late Mesozoic to recent remagnetization in eastern China (e.g., Kent et al., 1987; Zhao et al., 1990; Huang and Opdyke, 1991). In apparent contradiction with some geological observations in the Qinling mountains, which led Mattauer et al. (1985) to conclude that suturing between the NCB and SCB had to be over by the end of the Paleozoic, the large post-Permian rotation found by Zhao and Coe (1987) has been confirmed by new Triassic results from the Ordos basin in the NCB (Yang et al., 1991). However, the lack of reliable Jurassic data make it difficult to describe the accretion process throughout the Mesozoic. For as yet unclear reasons, Jurassic data have been notoriously difficult to obtain in Asia. The present study on Jurassic formations from the Ordos basin (NCB) was undertaken in order to try and fill in some of the missing constraints.

Geological Setting and Sampling

The Ordos Basin is a large Permian to early Cretaceous nonmarine interior sedimentary basin. The Jurassic formations outcrop in the center and east of the basin and in general consist of grey to green feldspathic sandstones, siltstones and sandy shales, interbedded with rich coal seams. The sequence grades upward into finer sediments and is subdivided into four formations, extending from the Lower to the Middle Jurassic, based on extensive paleontological data

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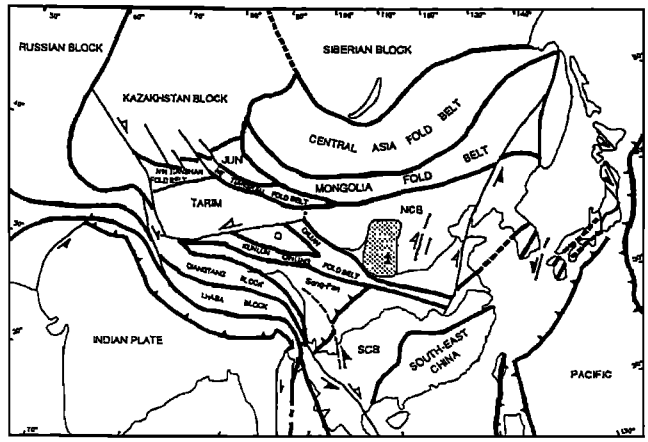


Fig.1. Tectonic sketch map of Asia (after Yang et al., 1991). The study area (Ordos basin on the NCB) is shaded. Sites (shown as triangle symbols) are at 36.2° N latitude and 109.3° E longitude. Arrows correspond to the sense of strike-slip motion on major faults (open for recent, closed for older phases).

(e.g. Wang, 1980). Jurassic sediments were sampled in the Yan'an area (Figure 1). They outcrop as a monoclinal sequence with dip less than 15°. Ten sites have been collected from the Lower Jurassic Fuxian Formation at Fuxian (red to grey sandstones), eight sites from the Lower/Middle Jurassic Yan'an Formation (sandstones, mud and siltstones, shales) and four sites from the Middle Jurassic Zhiluo (green to grey sandstone, mud and siltstones) and Anding (grey to red mudstones, shales, siltstones) Formations at Yan'an. A total of 182 cores were drilled, using a portable gasoline-powered drill, and oriented with a magnetic compass and occasionally with a sun compass.

Paleomagnetic Results

The Middle Jurassic and part of the Lower Jurassic samples were measured in the Beijing laboratory with a Digico and a Schonstedt DSM-2 magnetometer. All specimens were demagnetized by heating. At least 12 steps of thermal demagnetization were carried out up to 680° C. Part of the Lower Jurassic specimens were measured at the Paris laboratory, using a three axis CTF cryogenic magnetometer, Schonstedt GSD-1 alternating field demagnetizer and laboratory-built furnace. All data were analyzed in Paris using principal component analysis (PCA) (Kirschvink, 1980) and site means of directions and great circles were determined with the method of McFadden and McElhinny (1988).

Only six out of the 12 Middle Jurassic sites yielded internally consistent data. Rejected sites generally had weak magnetization and were unstable upon heating. These sites usually had a low unblocking temperature component around 200° C, with a direction corresponding to the present earth field (PEF) direction. The high-temperature components defined by PCA were associated with unacceptably high mean angular deviation values (greater than 20°) and magnetic mineral changes indicated by susceptibility and colour

changes were found in samples at some sites. The six acceptable sites show at least two components, a low-temperature component (100° to 150°C) close to the PEF direction and a high-temperature component (300° to 600°C) yielding internally consistent directions (Figure 2a).

Interestingly, site JAY1 shows a low-temperature *reversed* component (up to 100-150°C) and a high-temperature *normal* stable component (Figure 2a). The same phenomenon was observed in some Middle-Jurassic rocks from the Datong area, Shanxi province, 500 km away (Zheng et al. 1991). Zheng et al. suggest that the low-temperature component may be carried by goethite and that the high temperature component is carried by hematite and is primary. Site JAY2 (Figure 2a) is predominantly reversed from the NRM to the final demagnetization step; the Zijderveld diagrams often showed a flattened loop from about 150° to 300° C, suggesting an intermediate-temperature normal component, close to the PEF direction and opposite to the main characteristic direction. The carrier of the low-temperature (reversed) component could be goethite, and may have formed in recent time, although an ancient remagnetization is not ruled out by the data. The four other sites show a stable component up to 400°C, after which specimen intensities are close to the noise level of the magnetometer. These sites give internally consistent high-temperature components.

The characteristic directions for the six sites and their mean are listed in Table 1 and shown in Figure 3a. Half of the sites are normal and half are reversed. The directions are antipodal and can be averaged, yielding D=13°, I=56° (α95=4.5°) in geographic coordinates and D=14°, I=46° (α95=4.5°) after bedding correction.

Ten Lower Jurassic sites were demagnetized with both thermal and alternating field techniques. Isothermal Remanent Magnetization (IRM) experiments reveal the presence of a low-coercivity (around 10 mT) mineral, probably magnetite, in most specimens. Some IRM acquisition curves reveal the presence of a minor high-coercivity remanence carrier, probably haematite or goethite. Many specimens showed three components (Figure 2b). The first component is generally in a random direction and was possibly acquired in the laboratory; the second low-coercivity or lower-temperature component corresponds to the PEF direction. After 50 mT or 150° C a stable component was isolated (Figure 2b). In some cases only great circle

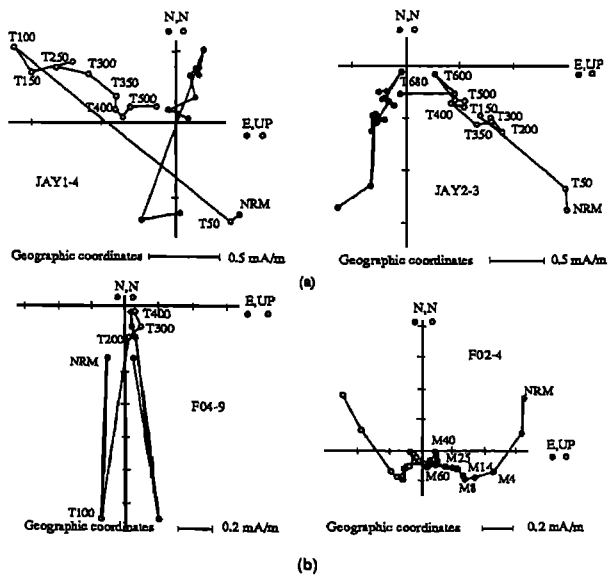


Fig.2. a). Orthogonal projections of thermal (T, °C) and alternating field (M, mT) demagnetization data of representative samples from a Middle Jurassic and b) Lower Jurassic sites (geographic coordinates). Solid circles (resp. open circles) refer to projections on the horizontal (resp. NS vertical) plane.

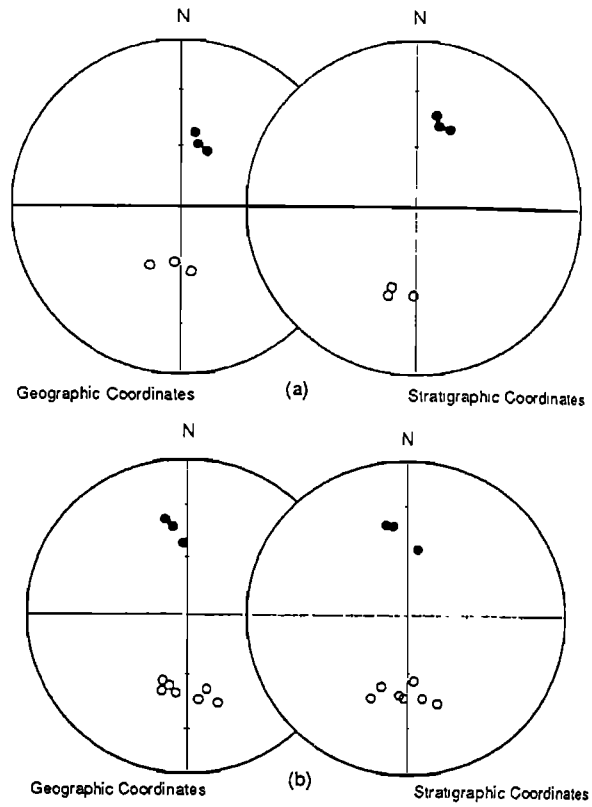


Fig.3. Characteristic site mean directions plotted in stereographic projection, before and after tilt correction. Solid (resp. open) circles represent positive (resp. negative) inclination. a) Middle Jurassic sites; b) Lower Jurassic sites.

constraints were obtained. Three sites are normal and seven are reversed (Table 1 and Figure 3b), and the respective means are statistically antipodal. The final mean direction is D=358°, I=47° (α95=6°) in geographical coordinates and D=0°, I=47° (α95=7°) in stratigraphic coordinates, which is barely distinct statistically from the PEF (dipolar PEF direction is D=0°, I=56°).

One may wonder whether the isolated magnetization obtained from this study might be an overprint, since it is quite close to the PEF, either before or after bedding correction. However, the reversal test for this higher temperature component and the magnetic behavior of the Middle Jurassic samples favour a primary magnetization. This point is addressed further below.

Discussion and Conclusion

We now compare our results with the few other published Jurassic studies known to us (Table 2). Our Middle Jurassic result compares reasonably with those of Zheng et al. (1991) from the Datong area and of Fang et al.(1988) from the Jingle area in Shanxi province. However, the small number of sites (only three) and the lack of reversed directions in both these studies were reason for concern. With six sites and a reversal test, we appear to be in a somewhat better position, although the fact that the published directions of Zheng et al. (1991) and Fang et al. (1988, no details on dip correction) agree better with our uncorrected directions is still a reason to be cautious.

Upper Jurassic results for the NCB have been reported by Lin (1984) and Cheng and Fang (1980), coming from five separate localities (Table 2). These results appear to be generally consistent, except for a mean direction reported by Cheng and Fang for the Wulian area, which is rotated by about 30° counterclockwise compared to the four other ones. This is based on only 5 sites and 13 specimens altogether. Moreover, these five studies fail to have recorded a single

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Table 1: Characteristic Jurassic directions from Yan'an area

Site	N(n,c)/N'	Dg	Ig	Ds	Is	k	α95
Lower Jurassic (36.16°N, 109.3°E)							
F1	5(3,2)/11	186.6	-48.5	186.4	-47.5	50.7	10.9
F2	5(3,2)/10	347.9	38.4	347.1	40.1	61.7	9.8
F3	5(2,3)/10	191.5	-50.7	183.8	-46.4	9.6	26.0
F4	5(2,3)/10	161.1	-39.7	162.4	-40.9	13.1	22.0
F5	4(3,1)/10	356.9	53.2	9.1	54.3	10.8	29.3
F6	5(5,0)/9	165.0	-48.6	170.7	-45.8	12.2	22.8
F7	5(3,2)/9	196.5	-47.2	204.3	-42.7	8.5	27.8
F8	4(2,2)/10	197.5	-52.7	200.0	-50.5	24.9	18.8
F9	6(6,0)/9	351.2	43.5	350.2	41.8	12.9	19.3
F10	6(5,1)/8	172.5	-44.0	176.0	-55.7	9.1	23.5
Mean	10	357.9	47.4	0.5	47.4	52.1	6.8
Mean nor.	3	351.6	45.1	354.3	45.8	53.5	17.0
Mean rev.	7	180.8	-48.2	183.2	-48.0	49.2	8.7
Middle Jurassic (36.7°N, 109.2°E)							
JYY3	6(3,3)/8	9.8	52.5	11.7	43.3	59.6	9.4
JYY4	7(1,6)/9	175.9	-53.6	182.0	-45.0	26.7	14.4
JYY5	7(1,6)/8	189.2	-58.3	198.4	-42.9	28.5	12.6
JZY	3(2,1)/9	23.4	60.6	22.4	48.1	128.0	10.9
JAY1	7(7,0)/11	13.6	58.1	15.1	48.2	63.9	7.6
JAY2	4(4,0)/9	207.1	-52.5	197.1	-47.1	135.1	7.9
Mean	6	13.0	56.4	14.4	45.9	220.8	4.5
Mean nor.	3	190.9	-55.5	192.5	-45.2	143.0	10.3
Mean rev.	3	15.1	57.2	16.3	46.6	298.5	7.1

Site numbers are given arranged in stratigraphic order, with oldest sites at top; N(N') are the number of retained (measured) directions or sites used in calculating the mean (n corresponds to stable endpoints and c to great circles); numbers of sites are indicated when means are given; Dg, Ig and Ds, Is: declination and inclination in geographic and stratigraphic coordinates respectively; k and α95: precision parameter and 95% angle of confidence.

Table 2: Mesozoic paleomagnetic data for eastern Eurasia

Site (°N/°E)	Age	N	A95 or dp/dm	Plat.	Plong.	References
NCB						
35.9/119.4	K1(+)	5	12/17	69.0	200.9	Lin, 1984
35.0/108.0	K1	10	6/9	75.9	210.8	Ma et al. 1991
35.9/119.4	J3(+)	10	6/10	71.3	225.8	Lin, 1984
35.8/117.5	J3(+,x)	10	4/5	74.2	215.6	Cheng and Fang, 1980
36.6/117.9	J3(+,x)	4	7/9	72.1	202.0	Lin, 1984
35.8/119.2	J3(+,x)	5	9/13	81.0	31.1	Cheng and Fang, 1980
31.5/116.5	J3(+,x)	5	4/7	72.7	229.4	Cheng and Fang, 1980
38.3/112.1	J2(x)	3	12/18	76.1	214.6	Fang et al., 1988
40.1/112.9	J2(x)	3	8	76.2	199.9	Zheng et al.,1991
36.7/109.2	J2	6	4/6	74.3	232.8	This study
36.2/109.3	J1	10	8/12	82.4	286.0	This study
35.2/109.2	T3	11	4	62.3	7.7	Yang et al. 1991
SCB						
30-26/103-120	K1	4	6	75.1	214.1	Enkin et al. 1991
29-26/101-120	J3	3	14	72.3	208.6	Enkin et al. 1991
27-26/101-106	J2	2	12	65.0	186.2	Enkin et al. 1991
26.5/102.2	J1	1	7	69.4	192.4	Enkin et al. 1991
26-32/101-119	T2-3	5	15	50.1	196.6	Enkin et al. 1991
Mongol-Central Asian Block/Fold Belts						
45.4/107.6	K1	3	5/6	82.9	221.7	Pruner, 1987
42.0/119.2	K1	6	6	82.9	249.5	Zhao et al., 1990
49.5/117.5	J3	24	9	68.5	231.6	Zhao et al., 1990

Column headings: Site latitudes and longitutes are given; Age ranges from upper Triassic (T3), through lower to upper Jurassic (J1 to J3) to Lower Cretaceous (K1); N number of sites (or individual studies for the SCB); A95 (or dp/dm) is 95% cone of confidence (or axes of ellipse of uncertainty); Other symbols: (+) Shandong sites not used in computation of mean because of suspected local rotation; (x) not used in mean because of either lack of detailed information or too few sites or samples.

reversal, despite a reversal rate of at least twice per million years in the period from 200 to 150 Ma. Also, there are no stability tests available. Finally, all sites are located rather close to the Tancheng-Lujiang fault system in Shandong province, and subsequent rotations and/or remagnetization in this general vicinity remain a possibility (Courtillot and Besse, 1986; Kent et al., 1987; Ma et al., 1991). As a consequence, these Upper Jurassic data can only be viewed with caution and are not used further.

Other available poles for the NCB from Upper Triassic to Lower Cretaceous are listed in Table 2 and tentative mean poles are displayed in Figure 4a.

One notes that the Lower and Middle Jurassic poles derived in this study plot nicely between the Triassic and Cretaceous poles. Indeed, a reasonable APWP linking the reliable Upper Triassic and Lower Cretaceous poles will pass close to the present North pole, thus yielding Jurassic directions close to the PEF. For this reason and for those discussed above, we believe that the higher temperature components which we have isolated may be primary. This feature of the NCB APWP may be one of the reasons for the difficulty in isolating Jurassic components from recent remagnetizations. The lack of significant folding prevents a fold test; sampling folded formations in the NCB remains a desirable goal.

Coeval mean poles for the SCB (Enkin et al, 1991) are plotted in Figure 4b. Comparison of Figures 4a and 4b shows that, whereas the NCB and SCB poles were still remote from their present relative positions in the Upper Triassic (see also Yang et al., 1991), they could have reached them in the Middle Jurassic since the two corresponding confidence intervals overlap. We see from Figure 4 that the NCB underwent little polar wander from the Middle Jurassic to early Cretaceous, but fast and large polar wander from the Upper Triassic to the Middle Jurassic (with an along-track rotation velocity of 0.8°/Ma). Polar wander of the SCB, however, was somewhat less during that time (Figure 4b).

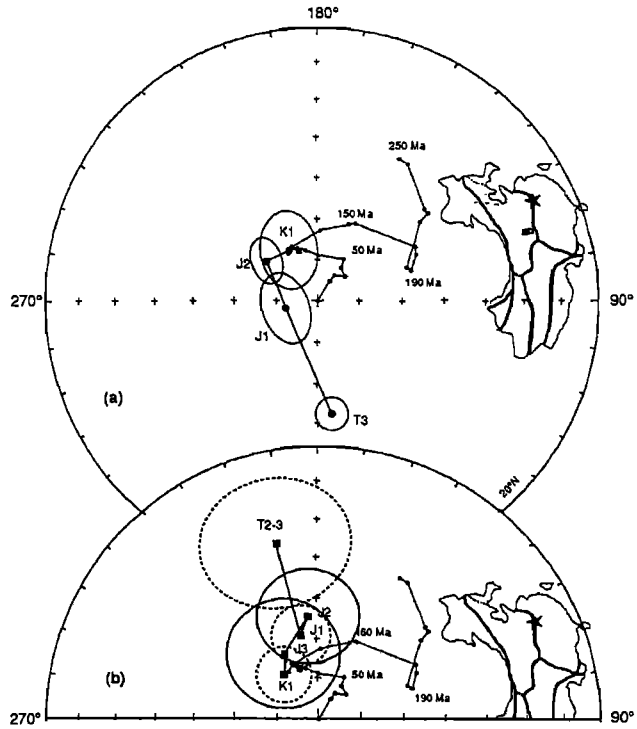


Fig.4. a) Apparent Polar Wander Path (APWP) of the North China block from the Upper Triassic to the Lower Cretaceous (solid circles). The synthetic APWP of the Eurasian plate since 250 Ma is given for reference (open circles) at 10 Ma intervals (Besse and Courtillot, 1991). b) APWP of the South China block for the same period (squares). For data and references see text and Table 2.

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The paleomagnetic data suggest that the NCB pole underwent a 47°+/-8° angular displacement between the late Triassic and the Middle Jurassic. This translates into a 76°+/-18° relative rotation between SCB and NCB for the same period, calculated at a reference point located at 31° N, 117° E (see star on Figure 4). This confirms the original suggestion of Zhao and Coe (1987), even emphasizing the total amplitude of the motion and setting more stringent time constraints. Such late major relative motion between the two blocks has been denied by several authors (e.g. Mattauer et al., 1985). However, there is some paleontological evidence that the late Triassic floras of parts of the NCB and SCB were quite distinct (Yancheng type vs. Shujianhe-Yipinglan type flora, see Yin, 1980). Evidence for significant motion and deformation during the Jurassic in the Qinling and Tongbai-Dabie mountains between the two blocks has been discussed by Zhao and Coe (1987) (see also Zhao et al., 1990; Enkin et al, 1991). Recent age determinations of thirty individual magmatic bodies (mostly granite) cluster around 230-220 Ma and 150-140 Ma (Ren et al., 1990). A granodiorite at Shagou (Shaanxi province) was reported at about 210 Ma (Reischmann et al., 1990). It has been suggested that part of the complex orogenic history of these ranges may correspond to an active continental margin. Pan (1989), for instance, placed the easternmost termination of the Paleo-Tethys Ocean between the NCB and SCB, in the Bayankala-Songpan area, and proposed that it had closed in the early Jurassic.

We see from data listed in Table 2 not only that the Middle/Late Jurassic poles of NCB and SCB are statistically similar (that of SCB having an unfortunately large uncertainty), but that they are also similar to those of Mongolia and the Central Asian ranges (Zhao et al., 1990). In any case, it appears that in the Middle/Late Jurassic, the SCB, NCB and Mongol blocks formed a single entity similar to the present day configuration but still remote from Siberia. Note that, for lack of direct data, the position of Siberia is estimated from the synthetic Eurasian APWP of Besse and Courtillot (1991). This favors the existence of a Mongol-Okhotsk ocean (e.g. Zhao et al., 1990) which had not yet closed in the Upper/Middle Jurassic. Some 2000 km of essentially North-South motion were still to take place north of Mongolia, until collision was complete probably in the early Cretaceous. The new Jurassic data reported here are further discussed in terms of paleogeographic reconstructions of the buildup of the Asian mosaic in a forthcoming paper (Enkin et al., 1991).

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